XAMS optical simulation: model, validation and first results

Geant4 optical simulation of the XAMS TPC with the meshes as real bars — geometry v2 from the drawings · 5 October 2026 · optical_sim/ (local git), user guide: optical_sim/docs (MkDocs)

Summary.

• Question: is the low S2 light on the top PMT a geometric effect? **Partly.** Geometry and optics give an S2 top fraction of 0.54–0.59 over all plausible optical parameters and liquid levels. The data without a source come close (0.50–0.68 depending on S2 size, 0.55 at 10–20 kPE); the Cs-137 and Na-22 values (0.27–0.40) are far below anything geometric — they are the load effect of the top PMT, not geometry.

• **The geometry is right:** the top-sector light pattern versus position (asymmetries u, v) lies inside the narrow simulation band over ±24 mm, identically for every source.

• S1 area fraction top versus depth has the shape of the simulation but a constant top/bottom ratio of 0.70 relative to it — one efficiency factor (small-signal integration on the top anodes and/or QE ratio), not geometry.

• The S2 area fraction top in the data dips at the centre (r < 5 mm); the model does not. Candidate: an insensitive cross between the four anodes of the top PMT.

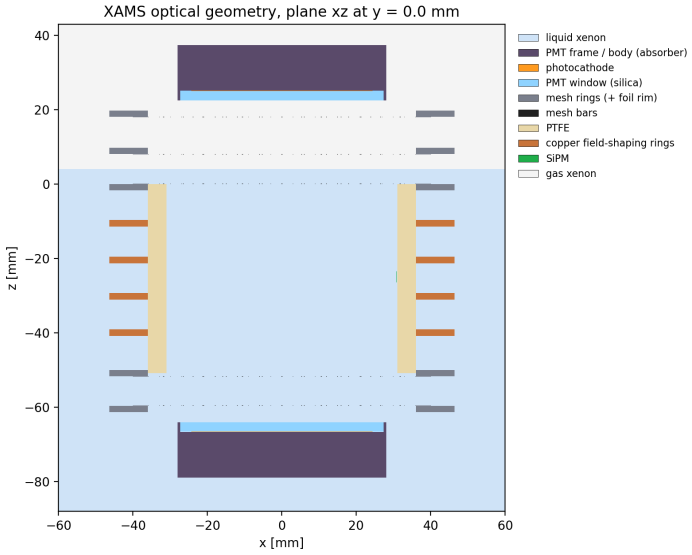
• Only ~23 % of the S2 photons reach a photocathode (QE 0.33 → 7.6 % detected); ~39 % go sideways out of the gate–anode / anode–top-screen gaps into the rings or out of the model. The liquid level (2 → 4 mm) hardly matters.

• Design study: PTFE cylinders also in the gate–anode and anode–top-screen gaps raise the S2 light collection by +36 %, with the cathode–bottom-screen gap +42 % (S1 +27 %) — a factor ~1.4, not more.

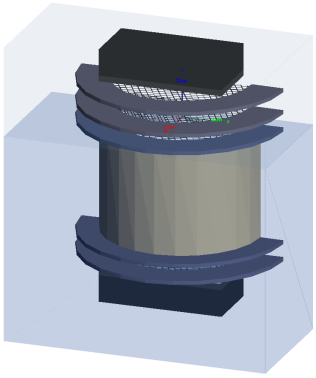
1. Model

One event = one 178 nm photon, started uniformly in the source volume with a random direction and polarisation and followed until it is detected or lost (G4OpticalPhysics: absorption, Rayleigh scattering, boundaries; no scintillation). S2: liquid surface to just below the anode foil, r < 30 mm; S1: cathode to gate, r < 30 mm. Every photon is recorded (origin, end point, how it ended, sensor, reflections); maps are made by binning the origins.

part	model (geometry v2, from the drawings)
meshes (5)	0.15 mm steel foils with square holes 2.45 mm, bars 0.15 mm (pitch 2.6 mm), all at 29.4° to the anodes, bars crossing on the axis (gate drawing); open aperture Ø72 mm. z: top screen +18, anode +8, gate 0, cathode −51.8, bottom screen −59.6 mm
mesh rings	steel, Ø72–92.6 mm, 1.8 mm assembly; the foil (Ø80) on one face, the ring body on the other side: below the gate and bottom screen, above the cathode, anode and top screen. The gate–anode gap is therefore open sideways between r = 36 and 46 mm
PTFE cylinder	Ø62/72 mm, 50.85 mm, in the bores of the gate and cathode rings (0.8 mm gap to the cathode foil); R 0.90–0.99, specular fraction 0–0.2, rest Lambertian
copper rings	four field-shaping rings Ø72–92.6 × 1.8 mm around the PTFE (8 mm gaps), R 0.2–0.4
xenon	liquid n = 1.69 up to the level (2 / 3 / 4 mm above the gate), Rayleigh 30–60 cm, absorption 1–10 m; gas n = 1.00 above
PMTs	two R12699: silica window 54.6 mm square, 2.5 mm, n = 1.60; photocathode 48.5 mm square with QE 0.33 (top: 4 sectors = ch1–ch4); frame and body absorbing
SiPMs	two, on the PTFE inner wall at z = −25 mm, azimuth 0° / 90°
outside	everything beyond |x|, |y| = 60 mm, below −90 or above +45 mm absorbs (to be made realistic)



Cross section at y = 0 from Geant4's own navigator (exactly what the simulation uses).



3-D view, cut open: PTFE (cream), mesh rings (grey), copper rings, PMTs.

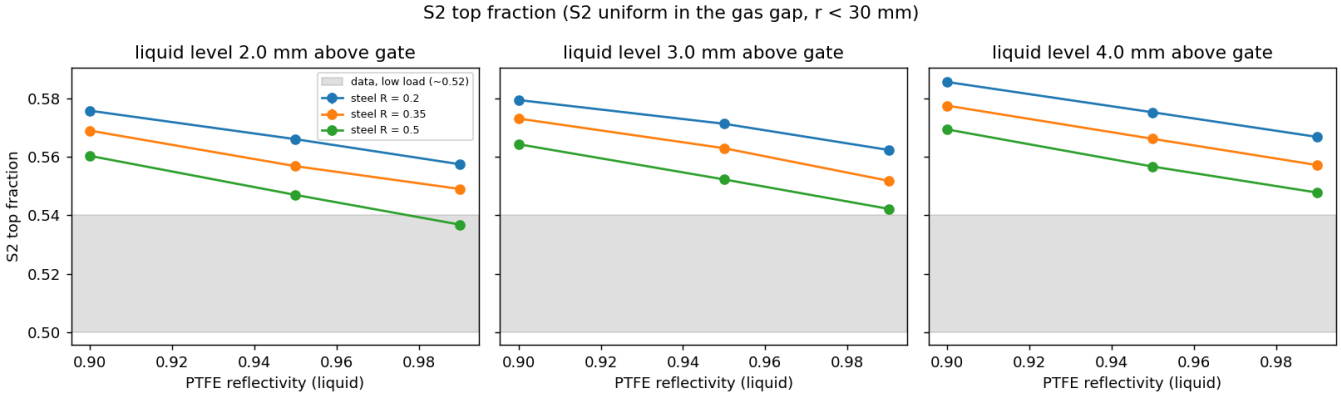
2. Validation

Each implementation step ended with a check against an analytic result:

check	result
solid angle (point source, 48.5 mm cathode at 10/30/60 mm)	detected fraction = solid angle, pulls ≤ 1.1
Fresnel + total internal reflection at the liquid surface, 0–80°	matches (Rs+Rp)/2; TIR from the critical angle 36.3°
mesh transparency, θ 0–75°, φ 0/29.4/45°	Geant4 = analytic ray trace photon by photon ($3 \cdot 10^6$ photons); 0.888 at normal incidence
optical surfaces (PTFE, steel)	reflected fraction = R, specular share = f, diffuse part Lambertian; R = 1 closes
sources and bookkeeping	z ranges follow the liquid level; every photon counted once
seed / statistics	independent halves of every scan point agree (worst pull 2.3)

3. Results

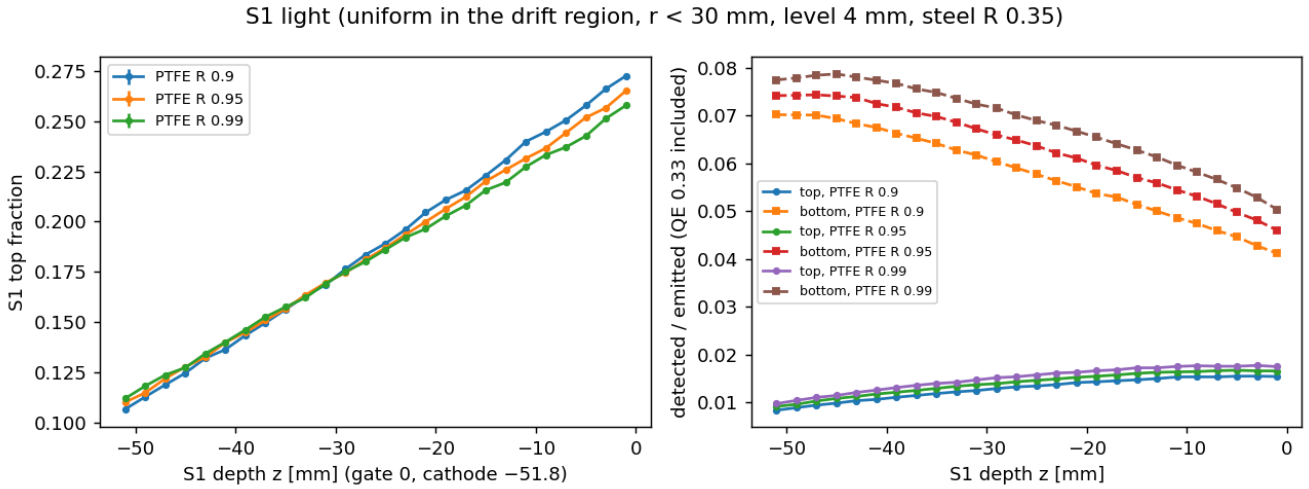
S2 top fraction. Over the full ranges of the free parameters and liquid levels: 0.537–0.586. Strongest knobs: PTFE specular fraction (−0.032 from 0 to 0.2), steel and PTFE reflectivity (≈ −0.02 each over their ranges), Rayleigh length (−0.013); liquid level 2 → 4 mm only +0.009. Versus radius 0.62 at the centre → 0.53 at the edge.



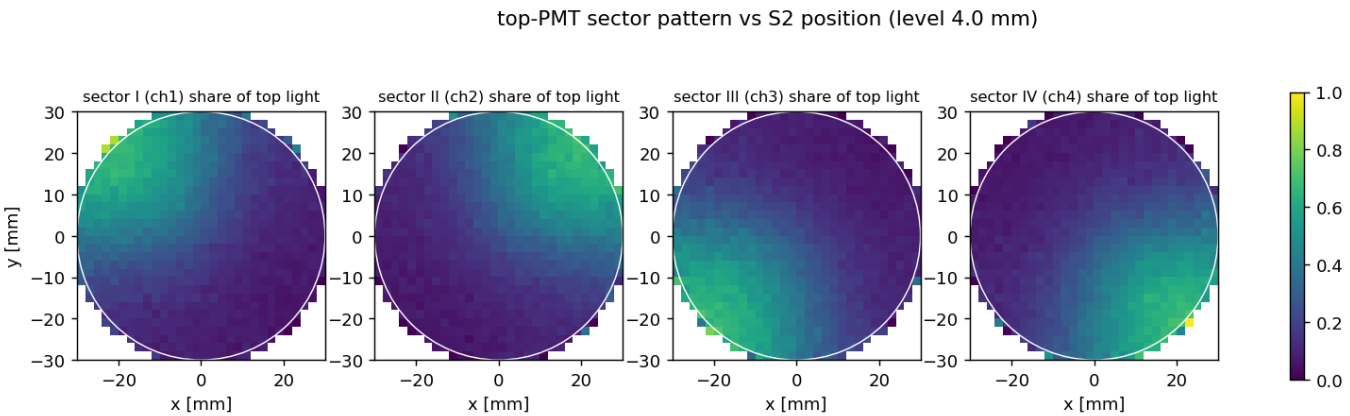
S2 top fraction vs PTFE reflectivity for three steel reflectivities and three liquid levels (10^7 photons per point; statistical errors ±0.0006).

Where the light ends. Reference point: level 4 mm, PTFE R 0.95, steel R 0.35, copper R 0.3.

	S2	S1
detected top / bottom PMT (QE incl.)	4.3 % / 3.3 %	1.4 % / 6.3 %
photocathode reached, not converted (1 − QE)	15.6 %	15.7 %
leaves the TPC region sideways	25.7 %	15.0 %
mesh rings + foil rims	13.2 %	11.6 %
mesh bars: anode / gate / top screen / cathode / bottom screen	10.4 % / 7.5 % / 4.0 % / 3.2 % / 2.2 %	1.9 % / 13.4 % / 1.1 % / 8.9 % / 5.2 %
PTFE	2.8 %	9.3 %
PMT frame / body	6.8 %	7.5 %
copper field-shaping rings	< 0.1 %	< 0.1 %
absorbed in the liquid	0.8 %	2.3 %



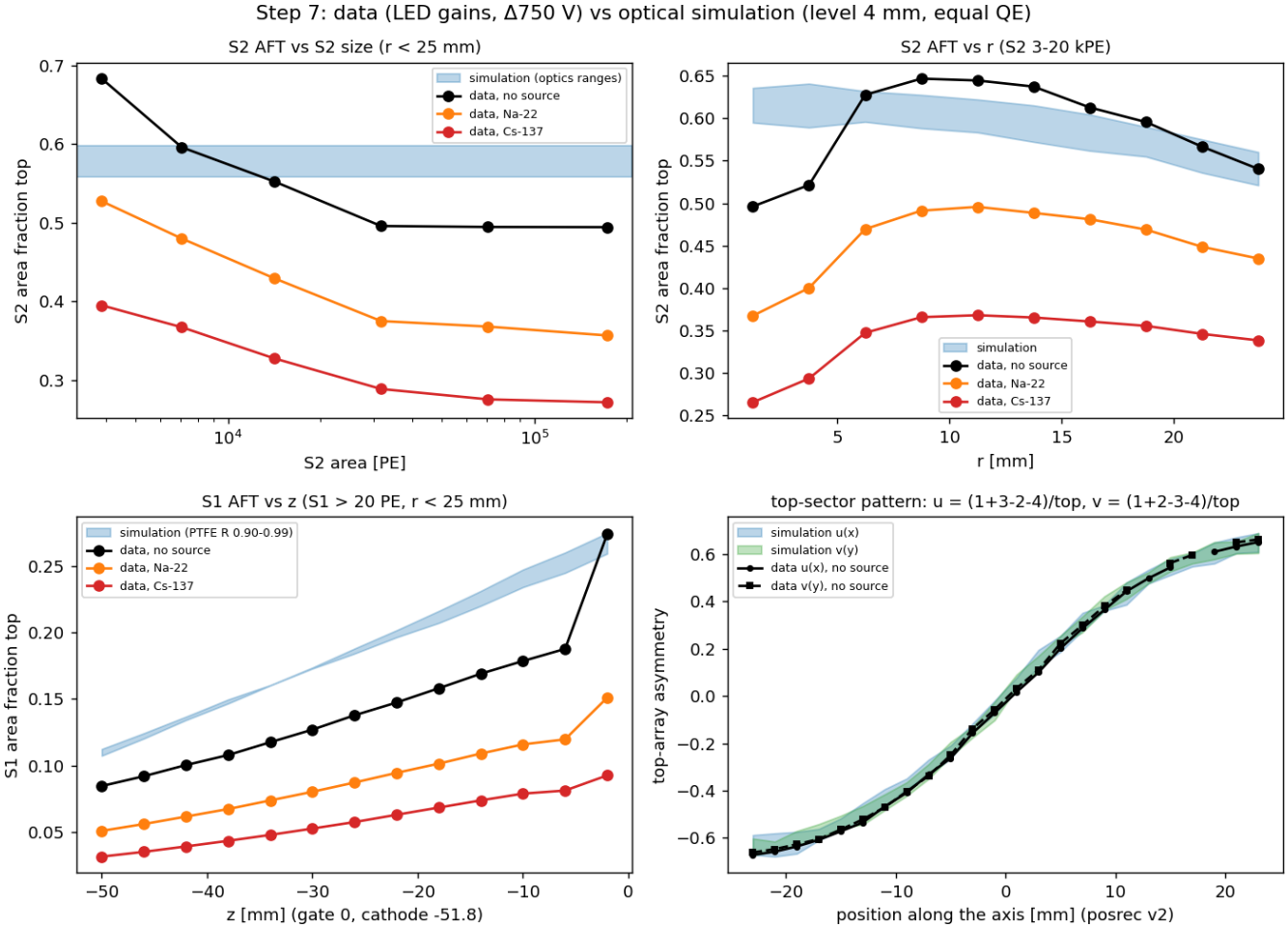
S1: top fraction (0.11 at the cathode → 0.26 at the gate) and light collection vs depth.



Share of each top-PMT sector in the top light versus S2 position (simulation frame: sector I at −x+y, II +x+y, III −x−y, IV +x−y).

4. Comparison with the data

Data: Δ750 V runs of 28/29-09 (central processing v4), no source / Na-22 / Cs-137, r < 25 mm, area fractions with the LED gains only (the v2 load correction f_top undone). Simulation band: all optics points at the 4 mm level, equal QE top and bottom.



Data vs simulation: S2 area fraction top vs S2 size and vs r, S1 vs z, and the top-sector asymmetries u(x), v(y).

• **Sector pattern:** data inside the simulation band over ±24 mm, identical for all sources — geometry right.
• **S2 versus size and source:** no source 0.68 (3–5 kPE) → 0.49 (> 100 kPE); Na-22 0.53 → 0.36; Cs-137 0.40 → 0.27. Rate and size dependence are top-PMT effects (load, large pulses), not geometry. The simulation (0.56–0.60) agrees with the low-load, medium-size data.
• **S2 versus r:** outside the centre data/simulation top/bottom ratio 1.0–1.2; at r < 5 mm 0.6–0.7 (central dip in every source).
• **S1 versus z:** same shape, constant ratio 0.70 (one efficiency factor for small signals on the top PMT).
• Implied top-PMT efficiency relative to geometry (S2): ≈ 1 without source, ≈ 0.3 for Cs-137 large S2 — the v2 gain factor f_top (0.74–0.88, anchored on Cs runs) corrects only part of it.

5. Design study: extra PTFE cylinders

PTFE cylinders with the radii of the drift-region PTFE, inside the ring bores, foil to foil:

extra PTFE	S2 light	S2 vs now	S1 light	S1 vs now	S2 top fraction
none (as built)	7.6 %	+0 %	7.7 %	+0 %	0.568
gate–anode	9.0 %	+17 %	8.3 %	+8 %	0.556
anode–top screen	8.9 %	+16 %	7.9 %	+2 %	0.592
cathode–bottom screen	8.0 %	+5 %	8.8 %	+14 %	0.542
gate–anode + anode–top screen	10.4 %	+36 %	8.5 %	+11 %	0.582
gate–anode + anode–top screen + cathode–bottom screen	10.9 %	+42 %	9.8 %	+27 %	0.560

Most rescued photons still end on meshes, PMT frames or in the photocathode (QE), so the gain is a factor ~1.4. Not studied: high voltage (PTFE across 5 kV gate–anode, charge-up in the extraction region), mechanics.

6. Limitations and next steps

• Everything outside |x|, |y| < 60 mm is black and the three gate–anode pillars are missing: ~26 % of the S2 light leaves the model sideways. The surrounding structure (holders, field-cage parts, cryostat wall) is next (TODO) — it affects light collection and the design study, little the top fraction.
• Fit (plan step 8): efficiency ratios top/bottom for S2 (QE) and S1 (QE × small-signal factor), with the optical parameters; test an insensitive cross on the top PMT for the S2 central dip.
• Oct 2024 (2 + 6 mm liquid/gas) vs now (4 + 4 mm): the simulation predicts only +0.01 in the S2 top fraction.